MARC FOLGADO BALBÁS

C++ / UNREAL ENGINE PROGRAMMER

Valencia, Spain | +34 656 368 537 | marcjoelfb@gmail.com | LinkedIn | Portfolio

C++ / Unreal Engine Programmer with nearly 2 years of combined experience in game development and production using Unreal Engine 5. Specialized in **runtime tools, animation systems and gameplay**, with hands-on experience in packaged builds, Control Rig and AI integration through MCP. Primarily targeting **Unreal Engine / Tools Programming** roles.

EXPERIENCE

Unreal Engine Programmer - MystherAI

Apr. 2026 - Present

Project-based collaboration

- Development of a **fully runtime animation editor** in Unreal Engine 5, with frame editing, bone selection and transformation, timeline workflows, and character manipulation tools.
- Design of the **C++ / Blueprints** architecture and evolution of the system toward **Control Rig**, ensuring full functionality in packaged builds without dependencies on editor-only tools.
- Development of reusable components, interfaces and animation systems to integrate editing tools directly into the application.

C++ / Unreal Engine Programmer - HechicerIA

Nov. 2025 - Apr. 2026

Project-based collaboration

- Development of an integration between Unreal Engine 5 and AI models through **Model Context Protocol (MCP)**, enabling engine functionality to be executed from model-generated requests.
- Implementation in C++ of **HTTP / WebSocket** communication, asynchronous workflows, and a dynamic tool registration and execution system.
- Development of runtime subsystems and interfaces designed for **packaged builds** and integration with external services.

Gameplay Programmer - Tiris / Cremaet Studios

Oct. 2024 - Jun. 2025

Game production · 9 months · Released on Steam

- Development of gameplay systems and mechanics throughout the full production of **Tiris**, a top-down shooter adventure later released on Steam.
- Implementation and maintenance of **enemy AI, inventory, pickups and Steam integration**, alongside debugging and gameplay system polish.
- Cross-disciplinary collaboration with programming, design and art throughout the production cycle, culminating in a release-ready build.

FEATURED TECHNICAL PROJECT

Raftel Engine - Custom Engine in C++ / OpenGL

2025

- Development from scratch of a game engine in C++ and OpenGL featuring **PBR rendering, dynamic lighting, shadows, HDR/Bloom, materials and OBJ mesh loading**.
- Creation of a scene editor using **ImGui**, with entity selection and manipulation, input systems and basic physics.
- Design and integration of the engine architecture connecting rendering, scenes, components and editing tools.

EDUCATION

HND in Video Game Programming - ESAT

2022 - 2025 · Valencia

Intensive hands-on training in **C++, Unreal Engine, Unity, graphics programming, artificial intelligence and engine architecture**, combining technical coursework with complete projects developed in multidisciplinary teams.

ESAT - #1 Spain · #3 Europe · #5 worldwide · GAMEducation Ranking 2026

Previously completed one year of Computer Engineering at Universitat Politècnica de València (UPV).

TECHNOLOGIES

C++ | **Unreal Engine 5** | **Blueprints**

UNREAL ENGINE

Control Rig · Animation Systems · Runtime Tools · Gameplay Systems · Behavior Trees · UMG · Slate · Packaged Builds

INTEGRATIONS

MCP · HTTP · WebSocket · Steam

OTHER

Unity · Kotlin · OpenGL · ImGui

TOOLS

Git · Perforce · SVN

LANGUAGES

Spanish	Native
Valencian	Native
English	Cambridge C1